

Pruthuraj Parikh

MSc Student · Frankfurt am Main · ML × Backend × Data
Open to Werkstudent roles · Up to 20 hrs/week

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ABOUT

MSc student in High Integrity Systems at Frankfurt University of Applied Sciences. Strong foundations in Python, ML pipelines, backend engineering, and data analytics. Building real end-to-end systems from scratch — including a VITS-based TTS model trained on 13,100 audio clips, REST APIs, and full data pipelines. Seeking a Werkstudent role in ML Engineering, Data Science, or Backend Development.

TECHNICAL SKILLS

Languages	Python (strong) · Java · JavaScript · C · C++ · C# · SQL
ML & Data	PyTorch · CUDA · Pandas · NumPy · Scikit-Learn · Matplotlib · Seaborn · TensorBoard
Backend & APIs	Flask · Django · REST APIs · Firebase Auth · BeautifulSoup · Scrapy
Databases	MongoDB · SQLite3 · MySQL · Firebase Realtime DB
Data Engineering	EDA · Feature Engineering · Time-Series Analysis · Mel Spectrograms · Pipeline Design
Frontend / Tools	React · TypeScript · HTML/CSS · Git · Postman · Android Studio · Figma

PROJECTS

HexTTs

2024 – Present

ML Engineer — VITS Text-to-Speech trained from scratch

ML Engineering

Stack: PyTorch · CUDA · VITS · LJSpeech · Mel Spectrogram · Griffin-Lim · TensorBoard · AMP

- Implemented VITS (Variational Inference TTS) end-to-end on LJSpeech — 13,100 audio clips, 45M parameter model trained on an RTX 3050 Ti (4GB VRAM).
- Built a full training pipeline with checkpoint resume, gradient clipping, AMP, and TensorBoard logging across all loss components (reconstruction, KL, duration).
- Engineered a mel-spectrogram caching system (`vits_data_cached.py`) that precomputes features to disk, achieving ~3× training speedup.
- Implemented per-epoch audio sample generation for live model quality monitoring without halting training.
- Built inference pipeline: text → phonemes (`g2p_en`) → mel spectrogram → WAV via Griffin-Lim vocoder.

ECG Digital Twin

2025 – Present

Safety-Critical ECG Monitoring & Simulation System

High Integrity Systems

Stack: MATLAB · Digital Twin · PTB-XL Dataset · Signal Processing · V-Model

- Designed a safety-critical ECG monitoring system following HIS principles — mode-based operation (NORMAL / DEGRADED / FAIL-SAFE), watchdog supervision, and deterministic alarm management.
- Built a physiological ECG core model with noise injection, QRS detection, HR/RR computation, and Signal Quality Index (SQI) gating for real clinical data ingestion (PTB-XL dataset).
- Implemented fault injection demonstrations and a full verification suite: unit tests, safety tests, and requirements-to-test traceability mapping following V-Model structure.
- Developed role-based GUIs (Physician, Nurse, Patient) with explicit event logging and test result export to Excel.

Datascope

11/2023 – 03/2024

Predictive Price Analytics Platform

Data Engineer & Predictive Analyst | API Developer & System Designer

Stack: Python · Flask · MongoDB · Pandas · NumPy · Web Scraping · Matplotlib · REST APIs

- Scraped large e-commerce datasets using automated Python scripts; cleaned, deduplicated, and normalized with Pandas.
- Designed Flask-based REST APIs for product comparison, price trend analysis, and baseline predictive modelling.
- Built modular backend architecture with MongoDB, CSV/XLSX export pipelines, and automated email notifications on price drops.
- Developed data visualization dashboards for price fluctuation trends and product comparison summaries.

WebReader

04/2024 – Present

Digital Library & User Behavior Platform

Data Pipeline Developer | System Architect & Core Developer

Stack: Python · Firebase Auth · SQLite3 · REST APIs · JavaScript

- Engineered offline-first storage with SQLite3 for downloaded media; integrated Firebase Auth with secure token handling.
- Built backend services for real-time book updates, TTS processing, and content delivery with scalable modular design.
- Designed data flows tracking user interaction metrics (page activity, engagement, access patterns) and built internal dashboards for book popularity analytics.

EDUCATION

M.Sc. High Integrity Systems

2025 – Present

Frankfurt University of Applied Sciences · Frankfurt, Germany

Bachelor of Computer Applications (BCA)

2021 – 2024

CHARUSAT University · Gujarat, India

CGPA: 9.11 / 10

LANGUAGES · CERTIFICATIONS · INFO

LANGUAGES

English C1 · Fluent
German A1 · In Progress
Hindi Native
Gujarati Native

CERTIFICATIONS

Data Science Teachnook
Software Conceptual Design NPTEL
Prompt Engineering for ChatGPT CHARUSAT

ADDITIONAL INFO

→ 20 hrs/week (Werkstudent)
→ ML pipelines & model training
→ Backend systems & APIs
→ Data engineering & analytics
→ High-integrity software design